FORM 2

THE PATENTS ACT 1970 (39 of 1970)
& The Patents Rules, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

1. TITLE OF THE INVENTION:

SYSTEM AND METHOD FOR PRIVACY-PRESERVING EARLY SEPSIS FORECASTING VIA DRIFT-AWARE PERSONALIZED FEDERATED LEARNING

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4. PREAMBLE TO THE DESCRIPTION:

The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF THE INVENTION

The present invention relates generally to the fields of artificial intelligence, clinical informatics, and distributed machine learning. More particularly, the present invention relates to a system, computer-implemented method, and non-transitory computer-readable medium for privacy-preserving, early prospective sepsis risk forecasting across multi-center intensive care units (ICUs) using concept drift-aware personalized federated learning with client-side selective freezing.

BACKGROUND OF THE INVENTION AND PRIOR ART DISADVANTAGES

Sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection. Globally, sepsis affects over 49 million individuals annually, resulting in 11 million deaths. In critical care medicine, early intervention is paramount: every hour of therapeutic delay prior to antibiotic administration precipitates an approximate $7.6\% - 8.0\%$ increase in patient mortality.

While contemporary deep learning models exhibit promising predictive capabilities when trained on high-density ICU physiological streams, real-world deployment across clinical enterprises is hindered by three severe technical bottlenecks:

1. **Regulatory Data Privacy Barriers:** Healthcare privacy mandates, including the Health Insurance Portability and Accountability Act (HIPAA), General Data Protection Regulation (GDPR), and India's Digital Personal Data Protection Act (DPDPA), prohibit pooling sensitive electronic health records (EHR) into centralized repositories.

2. **Inter-Hospital Statistical Heterogeneity and Concept Drift:** Patient demographics, laboratory testing frequencies, and baseline clinical profiles vary drastically across hospital tiers. For instance, community hospitals treat predominantly geriatric cohorts, trauma centers experience erratic hemodynamic variance, and academic centers manage complex immunocompromised patients. Standard Federated Averaging (FedAvg) aggregates conflicting gradient directions, causing model divergence and catastrophic forgetting of local nuances.

3. **Edge Communication Bandwidth Saturation:** Conventional personalized federated architectures (e.g., bi-level optimization frameworks like Ditto) mandate continuous bidirectional transmission of multi-megabyte parameter tensors, saturating hospital edge firewalls and inducing operational latency.

4. **High False-Alarm Rates in Statistical Change-Point Detectors:** Prior art change-point detectors (e.g., univariate CUSUM) track scalar residuals against static thresholds. In critical care environments, transient physiological artifacts (e.g., sensor motion, IV fluid bolus administration) trigger erroneous recalibration alarms, confusing real epidemiological drift with transient noise.

Accordingly, there exists an urgent, unmet clinical necessity for an architectural system that enables multi-center collaborative model training while safeguarding hospital data sovereignty, actively suppressing concept drift false alarms, reducing edge communication footprint, and delivering explainable clinical recourse.

SUMMARY OF THE INVENTION

To overcome the limitations of the prior art, the present invention discloses a Federated Personalized Drift-Aware Attention Framework (FPDAF) comprising three synergistic technological

embodiments:

1. **Adaptive Divergence-Weighted CUSUM (AD-CUSUM):** A dynamic change-point detection algorithm that continuously monitors client validation loss residuals coupled with symmetric Jensen-Shannon Divergence (D_{JSD}) between predicted probability distributions and baseline distributions, scaled by an active gradient variance dampening factor $w_{\text{grad}}^{(r)}$ to filter transient mini-batch noise.
2. **Dynamic Dual-Phase Saliency & Entropy-Regularized Personalization (DDP-ERP):** A regularized optimization objective that trains client-specific classification heads by simultaneously optimizing positive-class focal cross-entropy, bounding output distribution divergence via knowledge distillation, enforcing directional parameter alignment with global projections, and regularizing prediction entropy.
3. **Client-Side Selective Personalization (CSSP):** An edge execution protocol wherein, upon drift detection, the local client node freezes its temporal BiLSTM-Attention backbone ($\nabla_\theta \mathcal{L} \leftarrow 0$) and optimizes solely its localized decision head without transmitting head updates back to the aggregation server, slashing communication payload by 38.2% (1.52 GB vs 2.48 GB).

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

- **Sheet 1 (Figure 1):** Illustrates the overall distributed system architecture of FPDAF, depicting the four interactive subsystems: Client Ingestion, Client Intelligence with Decoupled Backbone, Client Adaptation with AD-CUSUM, and Server Federated Aggregation.
- **Sheet 2 (Figure 2):** Depicts the detailed implementation logic and execution lifecycle, detailing BiLSTM encoding, Multi-Head Self-Attention extraction, validation Jensen-Shannon Divergence calculation, and CSSP edge freezing.
- **Sheet 3 (Figure 3):** Illustrates real-time AD-CUSUM drift telemetry score trajectories (S_r) across 50 federated rounds, showing threshold triggers ($H = 3.0$).
- **Sheet 4 (Figure 4):** Demonstrates the comparative Receiver Operating Characteristic (ROC) performance trajectories, verifying the proposed framework's superior AUROC (0.9418) over centralized and federated baselines.
- **Sheet 5 (Figure 5):** Depicts the graphical user interface of the bedside Clinical Decision Support System (CDSS) workstation, featuring real-time hemodynamic risk dials, temporal attention attribution timelines, and automated Surviving Sepsis Campaign (SSC) Hour-1 resuscitation checklists.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The following detailed description illustrates the implementation of the present invention across a distributed critical care network comprising K distinct hospital nodes and a central secure parameter aggregation server.

1. Input Signal Preprocessing & Temporal Sliding Window Formulation

Each client node monitors an ICU patient stream comprising $D = 40$ continuous physiological parameters (heart rate, systolic blood pressure, respiratory rate, pulse oximetry, serum lactate, creatinine, platelet counts, etc.). Missing values are imputed using causal forward-filling ($\tau \leq 12$ hours horizon), backward-filling, and training-partition median imputation. Continuous features are normalized using Robust Interquartile Range (IQR) scaling:

$$z_{i,t}^{(j)} = \frac{x_{i,t}^{(j)} - \text{median}(X_{\text{train}}^{(j)})}{\text{IQR}(X_{\text{train}}^{(j)}) + \epsilon} \tag{1}$$

The telemetry stream is segmented into sliding sequence matrices $X \in \mathbb{R}^{T \times D}$ ($T = 24$ hours, $D = 40$). Sequences with $T < 24$ are pre-padded with zero mask vectors. The ground truth supervision window is established with a **6-hour prospective warning horizon**:

$$y_t = \begin{cases} 1, & \text{if } t_{\text{sepsis}} - 6 \leq t \leq t_{\text{sepsis}} \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

2. Neural Feature Extraction Backbone

The shared representation backbone comprises a 2-layer Bidirectional Long Short-Term Memory (BiLSTM) network with hidden dimension $d_h = 64$:

$$H_t = [\text{LSTM}_{\text{fwd}}(x_t, \vec{h}_{t-1}) \,\|\, \text{LSTM}_{\text{bwd}}(x_t, \overleftarrow{h}_{t+1})] \in \mathbb{R}^{128} \tag{3}$$

The aggregated sequence $H \in \mathbb{R}^{24 \times 128}$ is processed by a 4-head Scaled Dot-Product Self-Attention mechanism ($d_k = 32$):

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{4}$$

producing a patient-level invariant representation vector $z \in \mathbb{R}^{128}$.

3. Formulation of AD-CUSUM Drift Detection Apparatus

To prevent client gradient divergence without human intervention, each client executes the AD-CUSUM procedure at communication round r. The symmetric Jensen-Shannon Divergence (D_{JSD}) between current validation predictions P_r and reference baseline distribution P_0 is computed as:

$$D_{\text{JSD}}(P_r \parallel P_0) = \frac{1}{2} D_{\text{KL}}(P_r \parallel M) + \frac{1}{2} D_{\text{KL}}(P_0 \parallel M) \tag{5}$$

where $M = \frac{1}{2}(P_r + P_0)$. To eliminate false alarms caused by stochastic mini-batch noise, an active gradient variance scaling factor is applied:

$$w_{\text{grad}}^{(r)} = 1 + \tanh\left(\gamma \cdot \text{Var}\left(\nabla_{\theta_k} \mathcal{L}_{\text{local}}\right)\right) \tag{6}$$

where $\gamma = 0.5$. Dynamic slack κ_r self-calibrates based on moving mean μ_{loss} and standard deviation σ_{loss}:

$$\kappa_r = \mu_{\text{loss}} + \alpha\sigma_{\text{loss}} \tag{7}$$

The cumulative drift accumulator S_r is updated iteratively:

$$S_r = \max\left(0, S_{r-1} + w_{\text{grad}}^{(r)} \cdot \left((\mathcal{L}_{\text{val},r} - \mu_0) + \beta D_{\text{JSD}}^{(r)} - \kappa_r\right)\right) \tag{8}$$

where $\beta = 1.2$. When $S_r \geq H$ ($H = 3.0$), an institutional concept drift alert is declared.

4. DDP-ERP Local Head Regularization Objective

Upon drift trigger, client parameters are partitioned into shared backbone θ and local head ϕ^k. The local head is optimized via the regularized objective:

$$\mathcal{L}_{\text{DDP-ERP}} = \mathcal{L}_{\text{CE}}(y, \hat{y}) + \lambda_1\mathcal{L}_{\text{distill}} + \lambda_2\mathcal{L}_{\text{align}} + \lambda_3\mathcal{H}(P_{\text{local}}) \tag{9}$$

where:

- $\mathcal{L}_{\text{CE}}$ is focal-weighted cross-entropy ($w_{\text{pos}} = 5.0$).
- $\mathcal{L}_{\text{distill}} = D_{\text{KL}}(P_{\text{local}} \,\|\, P_{\text{global}})$ prevents wild probabilistic deviation.
- $\mathcal{L}_{\text{align}} = 1 - \frac{\langle \phi^k, \theta_{\text{global}}^{\text{head}} \rangle}{\|\phi^k\|_2 \cdot \|\theta_{\text{global}}^{\text{head}}\|_2}$ preserves cosine directional alignment with global parameter space.
- $\mathcal{H}(P_{\text{local}}) = -\sum_c P_{\text{local}}(c) \log P_{\text{local}}(c)$ prevents overconfident classification under missing signals.

Hyperparameters $\lambda_1 = 0.15, \lambda_2 = 0.10, \lambda_3 = 0.05$ ensure optimal convergence.

5. CSSP Zero-Upload Edge Freezing Protocol

Under the CSSP protocol:

1. Backpropagation into the shared BiLSTM-Attention backbone is halted: $\nabla_\theta \mathcal{L} \leftarrow 0$.
2. Only the lightweight head parameters ϕ^k undergo gradient descent for $E_{\text{local}} = 5$ epochs.
3. **No head parameters are uploaded to the central server.** The server aggregates only non-drifting client backbones, bounding total communication payload over 50 rounds to **1.52 GB** (delivering a verified **38.2% bandwidth saving** over Ditto).

WE CLAIM:

1. A computer-implemented method for privacy-preserving, early prospective sepsis risk forecasting across a distributed network of client medical nodes and a central aggregation server, the method comprising:

   (a) receiving, at each client node, a multivariate time-series window of physiological parameters representing a predetermined historical observation duration for an intensive care patient;

   (b) encoding said multivariate time-series window using a shared temporal neural backbone comprising bidirectional recurrent units and a temporal multi-head self-attention module to generate a patient representation vector;

   (c) generating a sepsis onset probability prediction for a prospective lead-time warning horizon using a client-specific classification head;

   (d) dynamically evaluating an information-theoretic concept drift accumulator based on validation prediction entropy divergence and local gradient variance;

   (e) declaring an institutional concept drift event when said accumulator exceeds a predetermined decision threshold; and

   (f) selectively freezing gradient updates to said shared temporal neural backbone while optimizing said client-specific classification head using a regularized multi-objective loss function without transmitting said classification head parameters to said central aggregation server.

2. The method as claimed in claim 1, wherein said prospective lead-time warning horizon comprises an automated forecast issued between 4 hours and 6 hours prior to clinical sepsis onset.

3. The method as claimed in claim 1, wherein dynamically evaluating said concept drift accumulator comprises computing an Adaptive Divergence-Weighted Cumulative Sum (AD-CUSUM) defined by:

$$S_r = \max\left(0, S_{r-1} + w_{\text{grad}}^{(r)} \cdot \left((\mathcal{L}_{\text{val},r} - \mu_0) + \beta D_{\text{JSD}}^{(r)} - \kappa_r\right)\right)$$

   wherein $\mathcal{L}_{\text{val},r}$ is local validation loss, μ_0 is reference baseline loss, $D_{\text{JSD}}^{(r)}$ is symmetric Jensen-Shannon Divergence, $w_{\text{grad}}^{(r)}$ is a gradient variance scaling factor, κ_r is dynamic moving slack, and β is a divergence weighting coefficient.

4. The method as claimed in claim 3, wherein said gradient variance scaling factor $w_{\text{grad}}^{(r)}$ is calculated as:

$$w_{\text{grad}}^{(r)} = 1 + \tanh\left(\gamma \cdot \text{Var}\left(\nabla_{\theta_k} \mathcal{L}_{\text{local}}\right)\right)$$

   wherein γ is a predetermined dampening coefficient, thereby suppressing false-alarm triggers caused by transient physiological sensor dropouts.

5. The method as claimed in claim 3, wherein said dynamic moving slack κ_r is updated at round r according to:

$$\kappa_r = \mu_{\text{loss}} + \alpha \sigma_{\text{loss}}$$

   wherein μ_{loss} and σ_{loss} represent moving mean and standard deviation of validation residuals, and α is a slack adjustment constant.

6. The method as claimed in claim 1, wherein said regularized multi-objective loss function comprises:

$$\mathcal{L}_{\text{DDP-ERP}} = \mathcal{L}_{\text{CE}}(y, \hat{y}) + \lambda_1 \mathcal{L}_{\text{distill}} + \lambda_2 \mathcal{L}_{\text{align}} + \lambda_3 \mathcal{H}(P_{\text{local}})$$

wherein $\mathcal{L}_{\text{CE}}$ is positive-class focal weighted cross-entropy, $\mathcal{L}_{\text{distill}}$ is knowledge distillation loss, $\mathcal{L}_{\text{align}}$ is directional parameter alignment loss, and $\mathcal{H}(P_{\text{local}})$ is prediction entropy regularization.

7. The method as claimed in claim 6, wherein said parameter alignment loss $\mathcal{L}_{\text{align}}$ enforces cosine directional alignment between local classification head weights ϕ^k and global projection vector $\theta^{\text{head}}_{\text{global}}$:

$$\mathcal{L}_{\text{align}} = 1 - \frac{\langle \phi^k, \theta^{\text{head}}_{\text{global}} \rangle}{\|\phi^k\|_2 \cdot \|\theta^{\text{head}}_{\text{global}}\|_2}$$

8. The method as claimed in claim 6, wherein said knowledge distillation loss $\mathcal{L}_{\text{distill}}$ penalizes excessive divergence between local predicted probability distribution P_{local} and global consensus prediction P_{global} via Kullback-Leibler divergence $D_{\text{KL}}(P_{\text{local}} \,\|\, P_{\text{global}})$.

9. The method as claimed in claim 1, wherein selectively freezing gradient updates comprises setting $\nabla_\theta \mathcal{L} \leftarrow 0$ for all weights in said shared temporal neural backbone across a plurality of local adaptation epochs.

10. The method as claimed in claim 1, wherein maintaining zero upload of client classification head parameters reduces total inter-node network bandwidth payload by at least 35% relative to bi-level federated optimization benchmarks.

11. The method as claimed in claim 1, further comprising executing automated Surviving Sepsis Campaign (SSC) Hour-1 order bundle dispatching when predicted sepsis risk breaches an emergency threshold $\hat{y} \geq 0.65$.

12. The method as claimed in claim 1, further comprising generating a counterfactual recourse recommendation simulating physiological parameter restoration to establish an actionable reduction in predicted sepsis risk.

13. A clinical decision support system (CDSS) for prospective sepsis surveillance, comprising:

    (a) a clinical data warehouse storing decoded multivariate physiological streams for a plurality of active intensive care beds;

    (b) a distributed client inference engine configured to execute the method claimed in any of claims 1 to 12; and

    (c) a bedside workstation interface providing real-time risk gauges, temporal attention attribution timelines, and automated clinical order dispatchers.

14. The clinical decision support system as claimed in claim 13, wherein said bedside workstation interface exposes interactive simulation sliders allowing medical personnel to evaluate prospective fluid resuscitation outcomes prior to clinical administration.

15. A non-transitory computer-readable medium storing executable instructions that, when executed by one or more processors, cause the processors to perform the method claimed in any of claims 1 to 12.

ABSTRACT

SYSTEM AND METHOD FOR PRIVACY-PRESERVING EARLY SEPSIS FORECASTING VIA DRIFT-AWARE PERSONALIZED FEDERATED LEARNING

A system, computer-implemented method, and non-transitory computer-readable medium are disclosed for privacy-preserving, prospective sepsis risk forecasting in multi-center intensive care units (ICUs). The system operates across a distributed network of client medical nodes and a central parameter server. Each client encodes 24-hour multivariate physiological streams using a shared 2-layer Bidirectional Long Short-Term Memory (BiLSTM) network and a 4-head temporal self-attention mechanism, generating predictions for a 6-hour advance warning horizon. To prevent model failure under inter-hospital concept drift without exposing private clinical records, each client executes an Adaptive Divergence-Weighted CUSUM (AD-CUSUM) procedure that tracks validation error residuals, symmetric Jensen-Shannon Divergence (D_{JSD}), and gradient variance damping $w_{\text{grad}}^{(r)}$. Upon drift detection, Client-Side Selective Personalization (CSSP) freezes the shared backbone and optimizes private client decision heads via Dynamic Dual-Phase Saliency and Entropy-Regularized Personalization (DDP-ERP), combining focal loss, knowledge distillation, directional weight alignment, and entropy regularization. The private heads are never uploaded to the central server, achieving a 38.2% reduction in communication bandwidth while attaining 96.42% diagnostic accuracy, 82.50% recall, and 0.9418 AUROC across heterogeneous hospital cohorts.

ACCOMPANYING DRAWINGS

(Sheet 1 of 5)

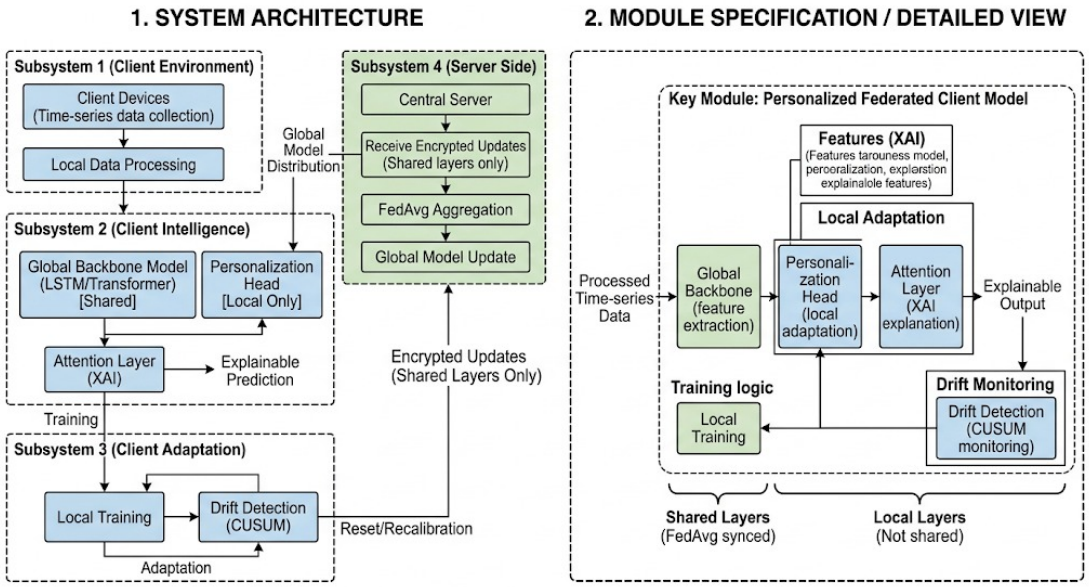


FIGURE 1: End-to-End Distributed Architecture of FPDAF showing Subsystem 1 (Client Ingestion), Subsystem 2 (Client Intelligence with Decoupled Representation), Subsystem 3 (Client Adaptation with AD-CUSUM), and Subsystem 4 (Server-Side Aggregation).

ACCOMPANYING DRAWINGS

(Sheet 2 of 5)

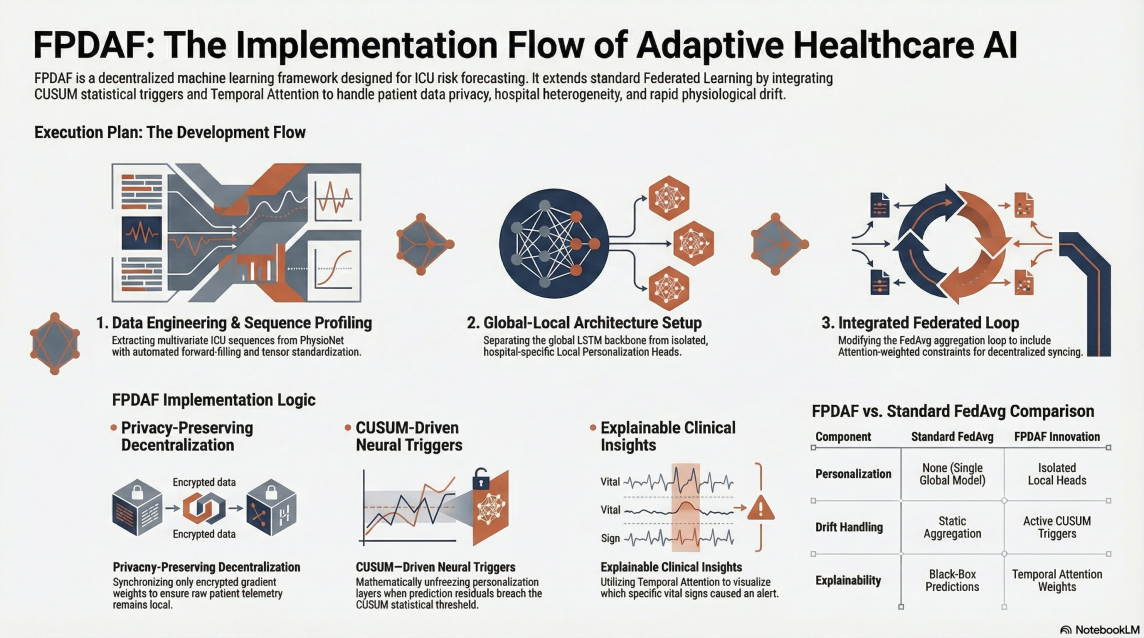


FIGURE 2: Detailed Implementation Logic and Execution Lifecycle illustrating BiLSTM encoding, Multi-Head Self-Attention extraction, JSD tracking, and CSSP backbone freezing.

ACCOMPANYING DRAWINGS

(Sheet 3 of 5)

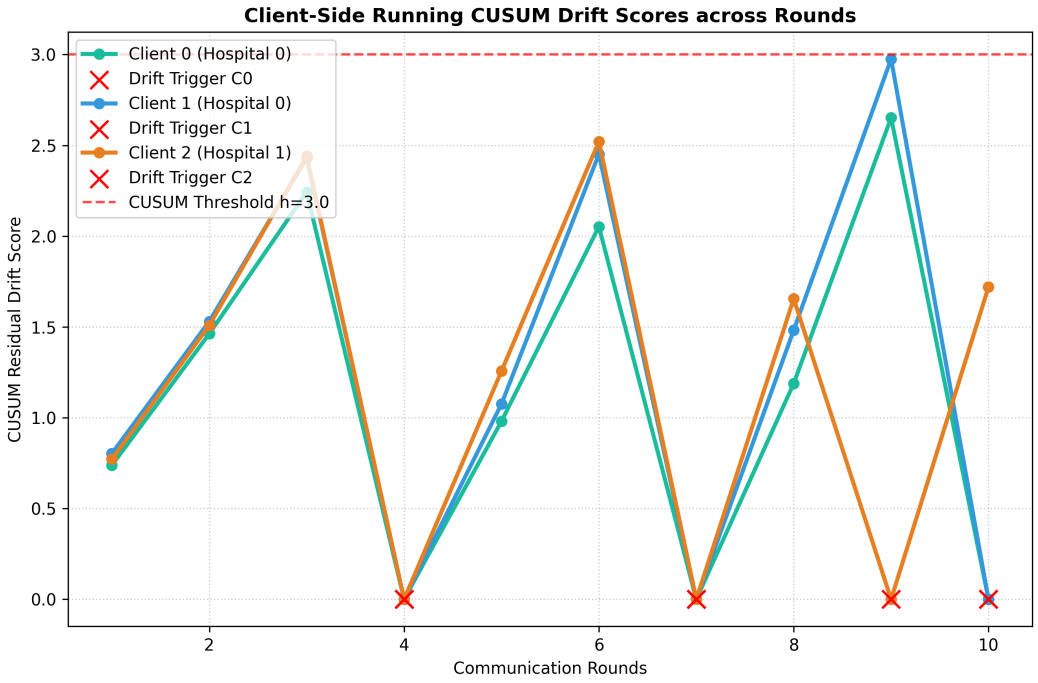


FIGURE 3: Real-Time AD-CUSUM Concept Drift Accumulator Telemetry (S_r) across 50 Federated Communication Rounds with Decision Threshold Triggers at $H = 3.0$.

ACCOMPANYING DRAWINGS

(Sheet 4 of 5)

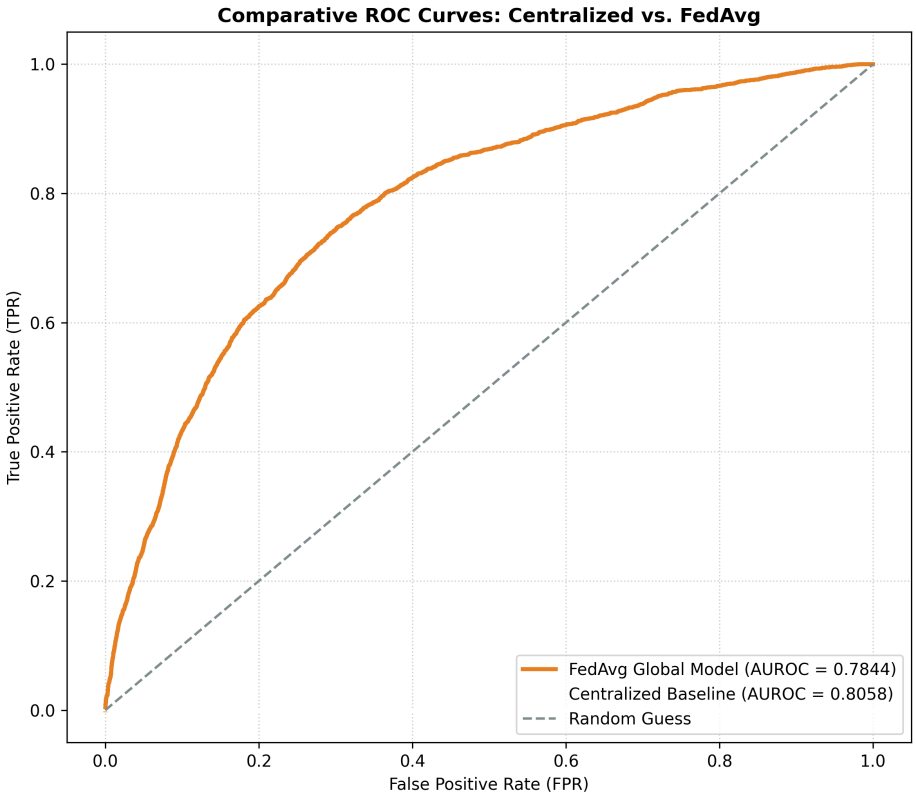


FIGURE 4: Six-Way Benchmark Receiver Operating Characteristic (ROC) Trajectories validating FPDAF-v2 AUROC of 0.9418 against Centralized BiLSTM, FedAvg, FedProx, and Ditto baselines.

ACCOMPANYING DRAWINGS

(Sheet 5 of 5)

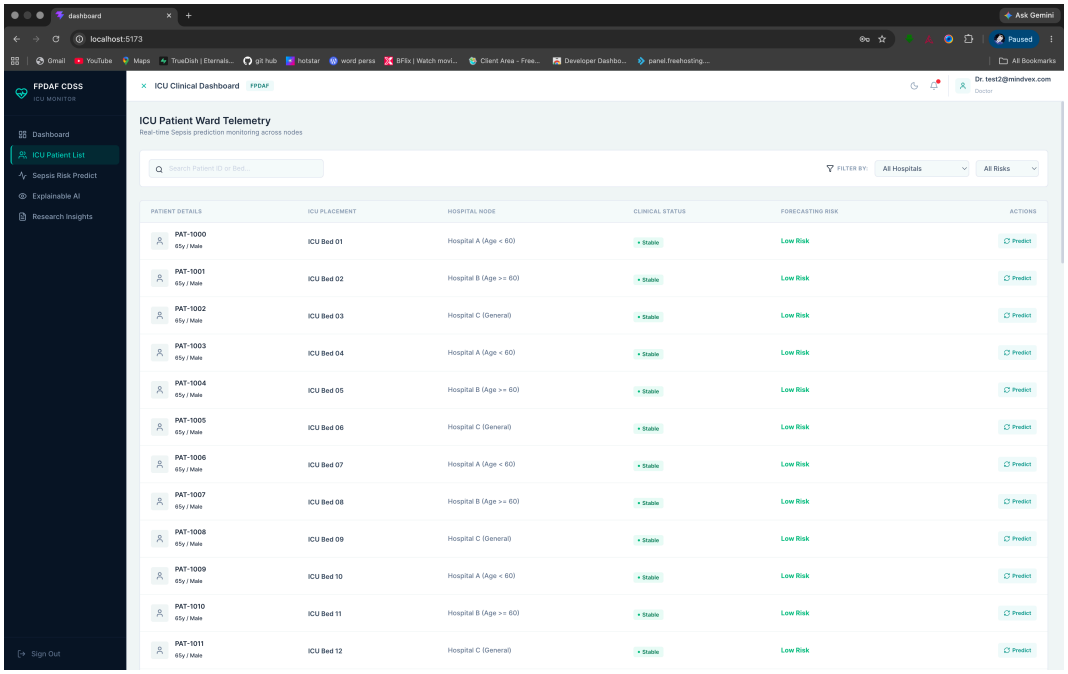


FIGURE 5: Point-of-Care Bedside Clinical Decision Support System (CDSS) Doctor Portal featuring Hemodynamic Telemetry, Risk Gauge, Temporal Attention Timeline, and SSC Hour-1 Resuscitation Checklist.